

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Mock Final  
 Duration: 95 minutes

Semester: Spring25  
 Full Marks: 40

**CSE 340: Computer Architecture**

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

- 1 a. **Convert**  $-111011.11011 \times 2^{12}$  in 26-bit IEEE-754 format where,
- i. Size of the fraction field is 13 bits. 2
  - ii. Size of the fraction field is 17 bits. 2
- Show** the equivalent Hex representation of your conversions. 1
- CO1 iii. Suppose you plan to use the converted number in a subsequent calculation where precision is crucial. Given a choice between the two IEEE formats (with 13-bit and 17-bit fraction fields), which format would provide a more accurate conversion for your calculation? Justify 1
- b. **Subtract** -6.55 from 15.21 using IEEE-754 single-precision floating-point representation. 2
- Consider** 5 decimal digits when you are converting from decimal to binary.
- c. For a 20-bit register with 1 bit for signs, 6 bits for exponents, and the rest for fractions, what will be the actual exponent range of the numbers that can be represented using IEEE 754 floating point representations? 1

- 2 a. Consider the below code sequence. Assume  $X_1$  has the value 0x00001232 and  $X_2$  has 1024 (decimal). Calculate the value  $X_3$  stores after executing each line of the code sequence. All the data values are in decimal notation in Table 2. Answer in Table 1 below and you must show all your calculations in your script. 3

| Code Sequence         | Value of $X_3$ |
|-----------------------|----------------|
| <i>lw X3, 8(X1)</i>   |                |
| <i>slli X3, X3, 2</i> |                |
| <i>Add X3, X3, X2</i> |                |

Table - 1

| Memory Address | Data (in decimal) |
|----------------|-------------------|
| 0x00001232     | 10                |
| 0x00001236     | 12                |
| 0x0000123A     | 13                |
| 0x0000123E     | 14                |

Table - 2

CO2

- b. Given a machine code 3  
 0x00CB1A93
- i. Find the RISC-V assembly code of the given machine code.
  - ii. Draw the datapath for the instruction you found in the previous question. You must mention the necessary control signals and proper labeling for all the components. 3

| Format | Instruction | Opcode  | Funct3 | Funct7/Funct 6 |
|--------|-------------|---------|--------|----------------|
| R      | ADD         | 0110011 | 001    | 0000000        |
|        | SUB         | 0110011 | 110    | 0000001        |
|        | AND         | 0110011 | 010    | 0000001        |
|        | OR          | 0110011 | 011    | 0000100        |
|        | SLL         | 0110011 | 101    | 0000100        |
| I      | LD          | 0000011 | 000    | N/A            |
|        | LW          | 0000011 | 010    | N/A            |
|        | LH          | 0000011 | 011    | N/A            |
|        | LB          | 0000011 | 001    | N/A            |
|        | ADDI        | 0010011 | 000    | N/A            |
|        | SLLI        | 0010011 | 001    | 0000000        |
|        | SRLI        | 0010011 | 010    | 0000000        |
|        | ORI         | 0010011 | 011    | N/A            |
| S      | SD          | 0100011 | 000    | N/A            |
|        | SW          | 0100011 | 001    | N/A            |
|        | SH          | 0100011 | 010    | N/A            |
|        | SB          | 0100011 | 011    | N/A            |

- 3 a. Explain why the Immediate Generation Unit requires the entire instruction for processing, rather than just the 12 bits of the immediate field? **3**
- b. Consider the code sequence given below.
- ld x<sub>10</sub>, 32(x<sub>11</sub>)  
add x<sub>5</sub>, x<sub>10</sub>, x<sub>7</sub>  
addi x<sub>5</sub>, x<sub>5</sub>, 3  
ld x<sub>13</sub>, 48(x<sub>5</sub>)  
sd x<sub>13</sub>, 32(x<sub>5</sub>)
- CO3
- i. Considering a single cycle datapath, find out how many clock cycles are required for the above code sequence and **calculate** the average CPI? **1+1**
- ii. How many data hazards are there in the given code sequence? **1**
- iii. Apply only **stall** to overcome the data hazards. **2.5**
- iii. Apply only **stall + forwarding** to overcome the data hazards. **2.5**
- iv. Calculate the total clock cycles and average CPI required after applying the method mention in the previous question(iii). **2**

- |                      |    |                                                                                                                                                                   |     |
|----------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4<br><br><br><br>CO4 | a. | Explain the significance of the offset field in an address with a proper example.                                                                                 | 1+1 |
|                      | b. | Explain why we need a memory hierarchy in our system.                                                                                                             | 2   |
|                      | c. | Find the cache index, Tag, and offset bits for the memory address = 328, given that the Block size is 8 bits and the cache size is 32 bits.                       | 1   |
|                      | d. | "Choosing a large size for the cache line helps minimize the cache miss penalty." Justify the validity of this statement with appropriate reasoning and examples. | 2+1 |
|                      | e. | Write a comparison between write-through and write-back techniques with proper examples.                                                                          | 1+1 |